ENGINEERS WALLAH

CHEMISTRY

a



Lecture V

Mass Number, Relative Masses and Percentage Composition

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TOPICS *to be covered*

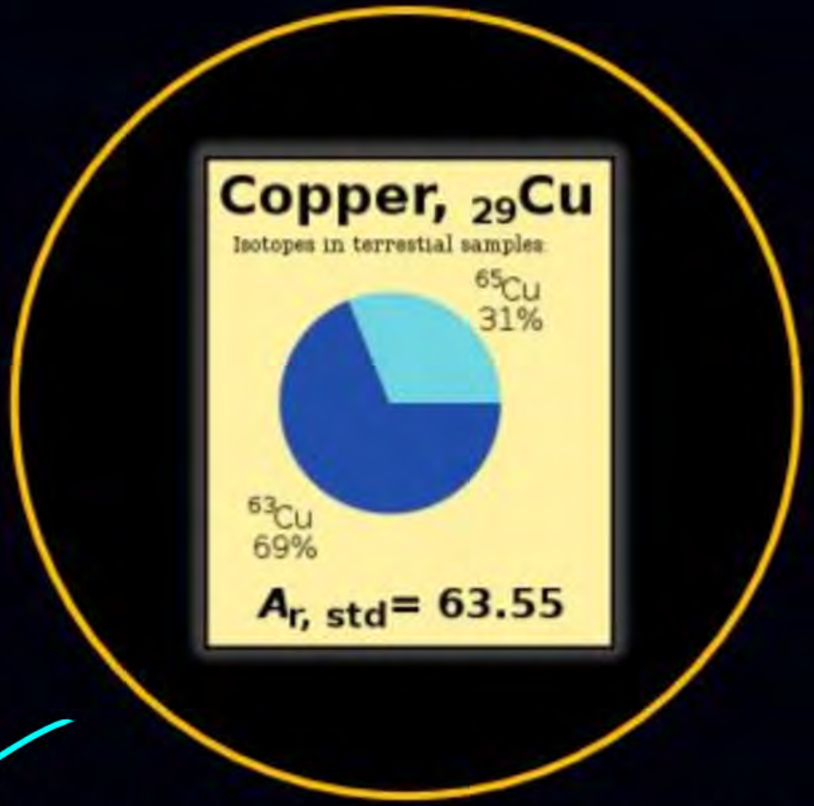
1. Mass Number (A)
2. Relative Atomic Mass, Molecular Mass and Formula Unit Mass
3. Percentage Composition of a Compound

Knowledge Ride On



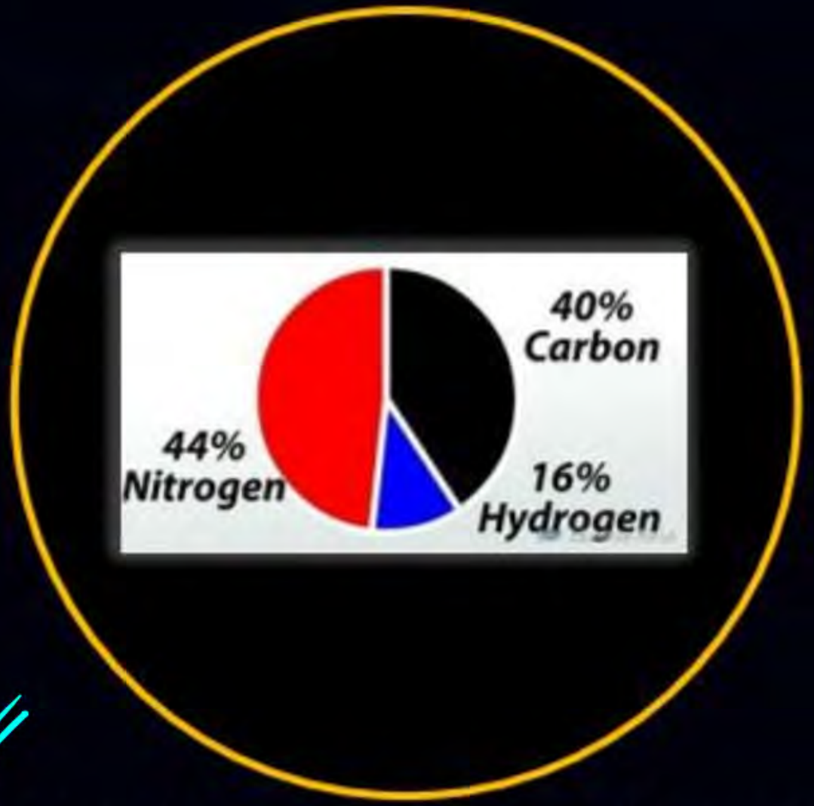
Mass Number (A)

Knowledge Ride On



Relative Atomic Mass, Molecular Mass and Formula Unit Mass

Knowledge Ride On



Percentage Composition of a Compound

Knowledge Ride On



Efficiency Hacks



Glossary

Mass Number (द्रव्यमान संख्या)
Atomic Mass (परमाणु द्रव्यमान)
Molecule (अणु)
Cation (धनायन)
Anion (ऋणायन)
Element (तत्व)
Covalent Compound (सहसंयोजक यौगिक)
Ionic Compound (आयनिक यौगिक)
Percentage Composition (प्रतिशत संरचना)

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

The charges on aluminium and sulphate ions are +3 and -2 respectively. The correct molecular formula of aluminium sulphate is

RRB JE 2015

_ _ _ _ _

A $AlSO_4$

B $Al_3(SO_4)_2$

C $Al_2(SO_4)_3$

D $Al(SO_4)_2$

Chemical symbol → Al SO_4

Charge → 3+ 2-

Valency → 3 2

$Al_2(SO_4)_3$

Aluminium sulphate

Concept Bites – Mass Number (A)

Mass Number (A), (द्रव्यमान संख्या)

C-I

C-II

(i) The total number of protons and neutrons present in the nucleus of an atom of element.

C-III

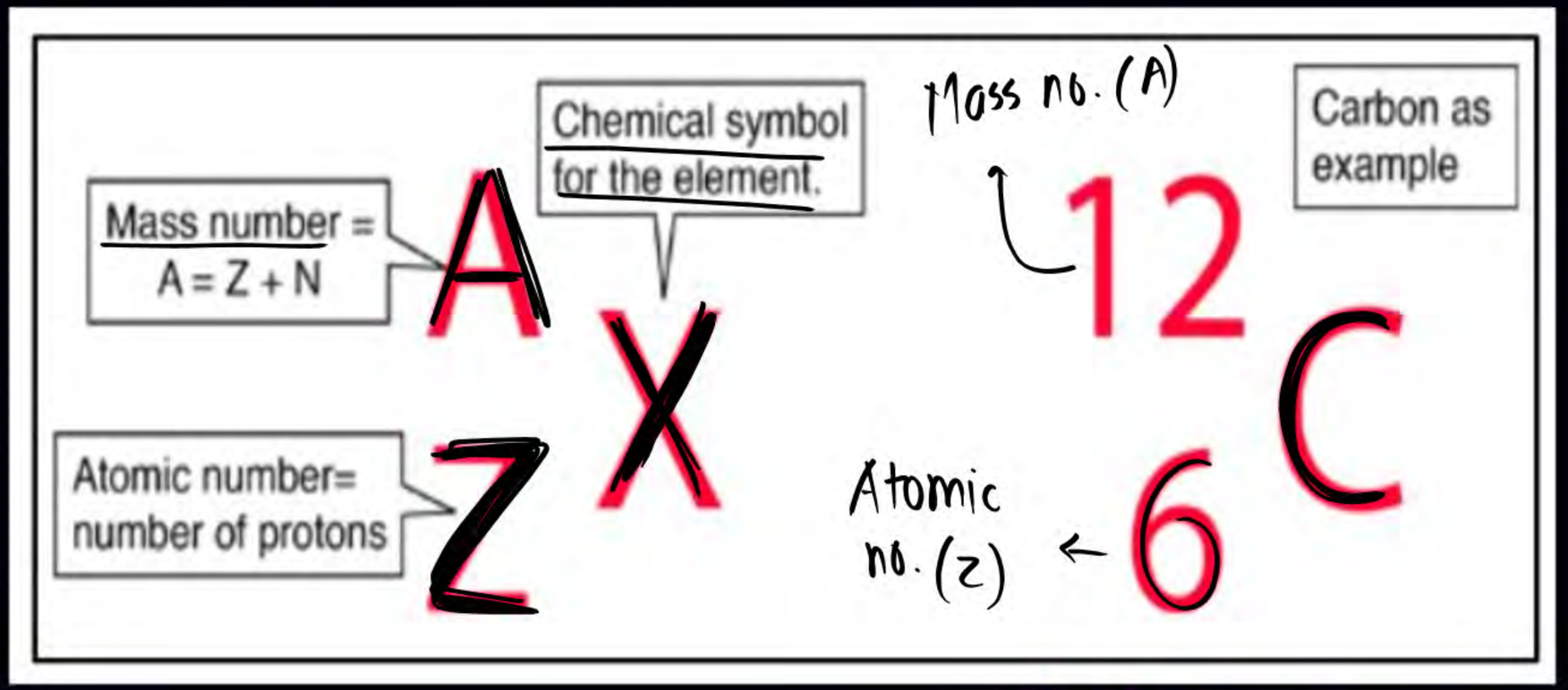
(ii) Mass number = Number of protons + Number of neutrons or Number of 'nucleons'

C-IV

(iii) It is always a whole number. → not fractional / decimal → 1, 2, 3, 4, 5 ...

1.5 (x), 2.6 (x) ...

Mass Number (A) - Representation



$A = Z + n$

$12 = 6 + n$

$n = 6$

Trick to Remember Mass Number of First 20 Elements

✓ Odd Atomic Number → 3, 5, 7, 9, 11, 13, 15, 17, 19 — 2Z+1

✓ Even Atomic Number → 2, 4, 6, 8, 10, 12, 14, 16, 20 — 2Z

Exceptions

'H or 'H'

$^{40}_{18}Ar$

Atomic no. 11
Sodium (Na)
→ (2×11)+1
→ 22+1
→ 23

$^{23}_{11}Na$

Atomic no. 6
Carbon (C)
→ 2×6
→ 12

$^{12}_{6}C$

Atomic no. 7
Nitrogen (N)
→ 2×7
→ 14

Atomic no. 13
Aluminium (Al)
→ (2×13)+1
→ 26+1
→ 27

Practice and PYQ's Wallah



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Question

$2Z+1 \rightarrow (2\times15)+1 \rightarrow 30+1 \rightarrow 31$

Which among the following is correct for the number of subatomic particles in an element X with atomic number (15) and mass number (31)?

परमाणु संख्या — द्रव्यमान संख्या

A. Protons: 15, Electrons: 15, Neutrons: 17

B. Protons: 16, Electrons: 16, Neutrons: 15

C. Protons: 15, Electrons: 15, Neutrons: 15

D. Protons: 15, Electrons: 15, Neutrons: 16

(I) Atomic no. (Z) → no. of protons = 15

(II) no. of protons = no. of electrons = 15

(III) $A = Z + n$

$31 = 15 + n$

$n = 16$

Question

The atomic number of an element X is 11 and it contains 12 neutrons in its nucleus. What will be the number of protons and the mass number of the element X?

Atomic no. (Z) = no. of protons = 11

A. Number of protons: 12, Mass number: 24 → 2Z + 1

B. Number of protons: 11, Mass number: 22 → (2 × 11) + 1

C. Number of protons: 12, Mass number: 23 → 22 + 1 → 23

D. Number of protons: 11, Mass number: 23

Question

An element X with atomic number 13 loses 3 electrons from its outermost shell. What will be the number of protons and mass number of the element X?

→ 'confuse you'

Atomic no. (z) = no. of protons = 13

A. Number of protons: 13, Mass number: 26

→ 27 + 1

B. Number of protons: 10, Mass number: 26

→ (2 × 13) + 1

C. Number of protons: 10, Mass number: 27

→ 26 + 1

D. Number of protons: 13, Mass number: 27

→ 27

Concept Bites – Relative Atomic Mass, Molecular Mass and Formula Unit Mass

Atomic Mass (परमाणु द्रव्यमान) - Meaning

Mass of an atom

(I) Mass of proton + neutron + electron + nuclear binding energy

electron + nuclear binding energy → negligible

(II) Modern Way → Mass Spectroscopy → To measure atomic mass.

Atomic Mass - Meaning

ATOMIC NUMBER	ELEMENT	ATOMIC MASS
1	Hydrogen	1.008
2	Helium	4.0026
3	Lithium	6.94
4	Beryllium	9.0122
5	Boron	10.81
6	Carbon	12.011
7	Nitrogen	14.007
8	Oxygen	15.999
9	Fluorine	18.998
10	Neon	20.180
11	Sodium	22.990
12	Magnesium	24.305
13	Aluminium	26.982
14	Silicon	28.085
15	Phosphorus	30.974
16	Sulfur	32.06
17	Chlorine	35.45
18	Argon	39.948

→ fractional / decimal values

Relative Atomic Mass

1803-05 → Dolton & Thomas Thomson

→ Atom is very small in size. It was difficult to measure its mass. Therefore, its mass was measured relatively.



$$\frac{(\text{mass})_{\text{sabu}}}{(\text{mass})_{\text{sunil}}} = 10$$

→ Initially, the reference chosen to measure the relative atomic masses of elements was hydrogen.



Relative Atomic Mass

After hydrogen, oxygen ($_8O^{16}$) was chosen as reference to measure the relative mass of atoms.

(International Union of Pure & applied Chemistry)

After oxygen, IUPAC in 1961 recommended carbon ($_6C^{12}$) as the reference to measure relative atomic masses.

Relative Atomic Mass (A_r) Values of Different Elements

Odd Atomic Number → 3, 5, 7, 9, 11, 13, 15, 19 — $2z+1$

Even Atomic Number → 2, 4, 6, (8), 10, 12, 14, 16, (20) — $2z$

A. No. (z) → 9

→ Fluorine (F)

→ $(2 \times 9) + 1$

→ $18 + 1$ → 19

Exceptions

H → 1

Ar → 40

Cl → 35.5

Imp.

S.I. unit of atomic mass → Kilogram (kg)

Non-S.I unit of atomic mass → unified mass (u) or Dalton (Da)

Unit of R.A.M (Relative Atomic Mass) → UNITLESS & DIMENSIONLESS

'Unitless'

$$\leftarrow \begin{cases} kg \leftarrow \dfrac{(mass)_{sabu}}{(mass)_{sunil}} = 10 \\ kg \leftarrow \end{cases}$$

Relative Molecular Mass (M_R)

The relative molecular mass (or molecular weight) of an element or a compound is sum of relative atomic masses of all elements present in that molecule.

(Covalent compounds)

$A_x B_y$

$$M_R = x \times \text{R.A.M of A} + y \times \text{R.A.M of B}$$

ex: H_2O

$M_R = 2 \times \text{R.A.M of H} + 1 \times \text{R.A.M of O}$

$= 2 \times 1 + 16 = 18$

Practice and PYQ's Wallah



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Question

What will be the molecular mass of sugar ($C_{12}H_{22}O_{11}$)?

A 352

B 342

C 362

D 372

M_R = 12 × R.A.M of C + 22 × R.A.M of H + 11 × R.A.M of O

= 12 × 12 + 22 × 1 + 11 × 16

= 144 + 22 + 176

= 342

Question

What will be the molecular mass of sulphuric acid (H_2SO_4)?

M_R = 2 × R.A.M of H + 1 × R.A.M of S + 4 × R.A.M of O

= 2 × 1 + 1 × 32 + 4 × 16

= 2 + 32 + 64

= 98

A. 98

B. 88

C. 108

D. 78

Question

What will be the molecular mass of S_8?

$M_R = 8 \times$ R.A.M of S

$= 8 \times 32$

$= 256$

A 256

B 128

C 300

D 296



Relative Formula Unit Mass

The relative formula unit mass of ~~an element or~~ a compound is sum of relative atomic masses of all elements present in that formula unit.

(ionic compound)

NaCl ⟶ F_R = 1 × R.A.M. of Na + 1 × R.A.M. of Cl

= 23 + 35.5

= 58.5

Practice and PYQ's Wallah



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Question

What will be the formula unit mass of calcium oxide (CaO)?

A 46

B 56

C 66

D 76

$FR = 1 \times R.A.M.\ of\ Ca + 1 \times R.A.M.\ of\ O$

$= 1 \times 40 + 1 \times 16$

$= 40 + 16$

$= 56$

Question

What will be the formula unit mass of copper sulphate ($CuSO_4$)?

R.A.M = 63.5

F_R = 1 × R.A.M. of Cu + 1 × R.A.M. of S + 4 × R.A.M. of O

= 1 × 63.5 + 1 × 32 + 4 × 16

= 63.5 + 32 + 64

= 159.5

A. 159.5

B. 249.5

C. 139.5

D. 169.5

Concept Bites – Percentage Composition of a Compound

$$\% = \frac{\text{Your marks}}{\text{Total marks}} \times 100\%$$

Percentage Composition of a Compound

It is the percentage by weight of each element in a compound.

% of hydrogen in water?

H_2O

$$\% \text{ of hydrogen in 1 molecule of } H_2O = \frac{\text{Mass of hydrogen in 1 molecule}}{\text{Total mass of 1 molecule}} \times 100\%$$

$$= \frac{2 \times 1}{(2 \times 1 + 16)} \times 100\% = \frac{200\%}{18} = \frac{100}{9} = 11.11\%$$

Practice and PYQ's Wallah



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Question

$NH_2 = 14 + 2 \times 1 = 14 + 2 = 16$

Calculate the percentage of nitrogen in urea (NH_2CONH_2)?

A 56.67%

B 46.67%

C 11.1%

D 12.1%

$$\% \text{ of nitrogen in urea} = \frac{\text{Mass of nitrogen}}{\text{Total mass}} \times 100\%$$

$$= \frac{28}{32 + 12 + 16} \times 100\%$$

$$= \frac{28}{60} \times 100\%$$

$$= \frac{140}{3}\% = 46.67\%$$

Question



Calculate the mass percentage of water in washing soda ($Na_2CO_3.10H_2O$) crystals?

$$\text{mass } \% = \frac{\text{mass of water}}{\text{Total mass}} \times 100\%$$

mass of water

$10(2\times1+16) = 180$

(A) 62.9%

(B) 63.9%

(C) 60.9%

(D) 58.9%

mass of $Na_2CO_3.10H_2O$ → $2\times23+12+3\times16+10(18)$

→ $46+12+48+180$

→ 286

→ $\frac{180}{286}\times100\% = 62.9\%$

Efficiency Hack!

Tricks to Remember Concepts

Make flashcards, flowcharts and mind-maps.




#sbsathhai
#pwsathhai
Thank
You !